

Encl 452 AS #3.

Q1) consider system with three task with Periods 10 ms, 39 ms & 1 s. If WCET is estimated at 4 ms, 12 ms & 98 ms respectively. What is total time-loading of system?
Is task guaranteed to have feasible schedule using RMS criterion? if not what would be the easiest rewrite that would make these task schedulable?

Ans. $T_0 = 10 \text{ ms}$ $WCET_0 = 4 \text{ ms}$
 $T_1 = 39 \text{ ms}$ $WCET_1 = 12 \text{ ms}$
 $T_2 = 1 \text{ s} = 1000 \text{ ms}$ $WCET_2 = 98 \text{ ms}$

$$U = \sum_{i=0}^2 \frac{C_i}{P_i}$$

$$= \frac{4}{10} + \frac{12}{39} + \frac{98}{1000}$$

$$= 0.4 + 0.30 + 0.098$$

$U = 0.798$

Here Threshold for RMS = $n(2^{1/n} - 1)$

$$= 3(2^{1/3} - 1) = 78\%$$

Here total utilization is (79.8%) > Threshold (78%) hence according to RMS criterion schedule is not feasible.

If we change time Period for Task 2 from 39 ms to 50 ms to make it a harmonic set

$$\text{Then, Utilization} = \frac{4}{10} + \frac{12}{50} + \frac{98}{1000} = 0.738$$

Here by increasing time Period to 50 ms instead of 39 ms we get a utilization < threshold hence schedule is feasible.

Q2 A system has 3 concurrent task described by table below:

Task	Cycle	Execution time	Priority	
Task A	10 ms	4 ms	3	$hP(A) = \phi$
Task B	20 ms	5 ms	1	$hP(B) = \{0, 1\}$
Task C	40 ms	10 ms	2	$hP(C) = \{0\}$
Idle	continuous	5 ms	...	

(i) Find system utilization.

$$U = \frac{4}{10} + \frac{5}{20} + \frac{10}{40} = 0.9$$

(ii) Here task Periods are a harmonic set so as long as $U < 100\%$ we have a feasible schedule.
in our case $0.9 < 100\%$. Task set is feasible.

(iii) Response time for each task.

lets assume R_0, R_1 & R_2 as response time for task A, C and B respectively.

Task A (highest priority) $\therefore R_0$: assume $R_0^{(0)} = 0$

$$\begin{aligned} \therefore R_0^{(i)} &= C_0 + \sum_{j \in hP(A)} \left\lceil \frac{R_0^{(i)}}{P_j} \right\rceil C_j \\ &= 4 \end{aligned}$$

$$\therefore R_0 = 4 \text{ ms.}$$

Task C

$$R_1 = 10 + \left\lceil \frac{0}{10} \right\rceil 4 = 10 \text{ ms}$$

$$R_1^1 = 10 + \left\lceil \frac{10}{10} \right\rceil 4 = 14 \text{ ms}$$

$$R_1^2 = 10 + \left\lceil \frac{14}{10} \right\rceil 4 = 18 \text{ ms}$$

$$R_1^3 = 10 + \left\lceil \frac{18}{10} \right\rceil 4 = 18 \text{ ms (converged)}$$

Task B (lowest priority) $\Rightarrow$ Assume $R_0 = 5$ as task B cannot respond before 5 ms.

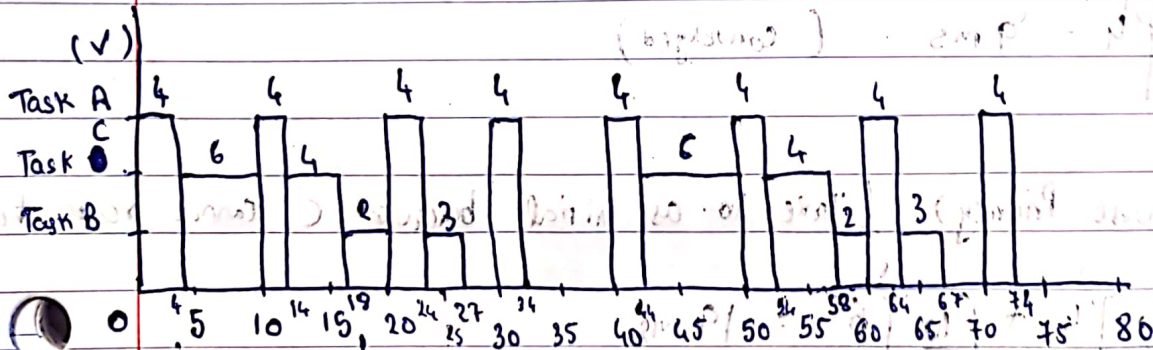
$$R_2^0 = 5 + \left\lceil \frac{5}{40} \right\rceil 10 + \left\lceil \frac{5}{10} \right\rceil 4 = 19 \text{ ms}$$

$$R_2^1 = 5 + \left\lceil \frac{19}{40} \right\rceil 10 + \left\lceil \frac{19}{10} \right\rceil 4 = 23 \text{ ms}$$

$$R_2^2 = 5 + \left\lceil \frac{23}{40} \right\rceil 10 + \left\lceil \frac{23}{10} \right\rceil 4 = 27 \text{ ms}$$

$$R_2^3 = 5 + \left\lceil \frac{27}{40} \right\rceil 10 + \left\lceil \frac{27}{10} \right\rceil 4 = 27 \text{ ms (converged)}$$

(iv) Here $R_0 = 4 < 10$ ms hence Task A meets deadline by 6 ms
 $R_1 = 18 < 40$ hence Task C meets deadline by 21 ms
 $R_2 = 27 > 20$ hence task B misses deadline by 7 ms.



(b) Now suppose Task B and Task C Priorities are changed.

New Priorities

Task A	3	R_0
Task B	2	R_1
Task C	1	R_2

(i) System Utilization = $\frac{4}{10} + \frac{5}{20} + \frac{10}{40} = 0.9$

(ii) ~~Response~~ Response time for each task

Task A: Highest Priority

$R_0 = 4 \text{ ms}$

Task B: (take 5 as initial because B cannot respond before 5ms)

$R_0^0 = 5 + \int_0^4 \frac{4}{10} = 9 \text{ ms}$

$R_1^1 = 5 + \int_5^9 \frac{4}{10} = 9 \text{ ms}$

$R_1^2 = 5 + \int_5^9 \frac{4}{10} = 9 \text{ ms} \quad (\text{converged})$

Task C: (lowest Priority) (take 10 as initial because C cannot respond before 10ms)

$R_2^0 = 10 + \int_{10}^{14} \frac{4}{10} + \int_{14}^{19} \frac{5}{20} = 19 \text{ ms}$

$$R_1^1 = 10 + \int_{10}^{19} 4 + \int_{20}^{19} 5 = 23 \text{ ms}$$

$$R_2^2 = 10 + \int_{10}^{23} 4 + \int_{20}^{23} 5 = 32 \text{ ms}$$

$$R_2^3 = 10 + \int_{10}^{32} 4 + \int_{20}^{32} 5 = 36 \text{ ms}$$

$$R_2^4 = 10 + \int_{10}^{36} 4 + \int_{20}^{36} 5 = 36 \text{ ms (converged)}$$

(iii) Do all task meet deadline.

Task A = 4 < 10ms meets deadline by 6ms

Task B = 19ms < 20ms meets deadline by 1ms

Task C = 36ms < 40ms meets deadline by 4ms

(iv) Draw an execution time line.

